

$$= \frac{47}{3} = 15\frac{2}{3}$$

174. According to given information average of odd numbers
= Average of even numbers - 6

$$\Rightarrow \frac{x_1 + x_2 + x_3 + x_4}{4}$$

$$= \frac{y_1 + y_2 + y_3}{3} - 6$$

$$\Rightarrow \frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{y_1 + y_2 + y_3 - 18}{3}$$

$$\Rightarrow 3(x_1 + x_2 + x_3 + x_4)$$

$$= 4(y_1 + y_2 + y_3) - 72$$

$$\text{Also, } y_1 + y_2 + y_3 = x_1 + x_2 + x_3 + x_4 - 16$$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 = y_1 + y_2 + y_3 + 16 \dots (i)$$

So, we have

$$3(y_1 + y_2 + y_3 + 16)$$

$$= 4(y_1 + y_2 + y_3) - 72$$

$$\Rightarrow 3y_1 + 3y_2 + 3y_3 + 48$$

$$\Rightarrow 4y_1 + 4y_2 + 4y_3 - 72$$

$$\Rightarrow 4y_1 + 4y_2 + 4y_3 - 3y_1 - 3y_2 - 3y_3 = 48 + 72$$

$$\Rightarrow y_1 + y_2 + y_3 = 120$$

$$\Rightarrow x_1 + x_2 + x_3 + x_4 = 120 + 16 = 136 \quad [\text{From (i)}]$$

$\therefore$ Average of four odd numbers

$$= \frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{136}{4} = 34$$

175. If a month begins with Sunday then there are 5 Sundays in that month.

Total number of visitors come on Sunday

$$= 510 \times 5 = 2550$$

Total number of visitors come on other days

$$= 240 \times 25 = 6000$$

$\therefore$ Average number of visitors per day

$$= \frac{2550 + 6000}{30} = \frac{8550}{30} = 285$$

176. Total marks of six boys = $48 + 59 + 87 + 37 + 78 + 57 = 366$

$$\text{Required average} = \frac{366}{6} = 61$$

177. Average weight of 75 girls = 47 kgs

$$\text{Total weight of 75 girls} = 47 \times 75 = 3525 \text{ kg}$$

Actual weight of 75 girls = x

Correct weight of 75 girls

$$3525 - 45 + 25 = 3525 - 20 = 3505 \text{ kg}$$

$$\therefore \text{Required average weight} = x = \frac{3505}{75} = 46.73 \text{ kg}$$

EXERCISE

(DATA SUFFICIENCY TYPE QUESTIONS)

Directions (Questions 1 to 28): Each of the questions given below consists of a statement and/or a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statement(s) is/are sufficient to answer the given question. Read both the statements and

Give answer (a) if the data in Statement I alone are sufficient to answer the question, while the data in Statement II alone are not sufficient to answer the question;

Give answer (b) if the data in Statement II alone are sufficient to answer the question, while the data in Statement I alone are not sufficient to answer the question;

Give answer (c) if the data either in Statement I or in Statement II alone are sufficient to answer the question;

Give answer (d) if the data even in both Statements I and II together are not sufficient to answer the question;

Give answer (e) if the data in both Statements I and II together are necessary to answer the question.

1. Sachin's monthly salary is ₹ 4,000. What is Rajan's monthly salary? (Bank P.O., 2010)

I. Rajan gets ₹ 500 more than the average salary of his and Sachin's.

II. Average of Sachin's and Rajan's salary is ₹ 4500.

2. Is r equal to the average of x , y and z ?

$$\text{I. } x + y + z = 3r \quad \text{II. } \frac{x + y + z}{6} = \frac{r}{2}$$

3. What marks have been obtained by Arun?

I. Arun's marks are the average of marks of Nitin and Manick.

II. Nitin obtained 80 marks and this is 25% more than the marks obtained by Manick.

4. Which of the four numbers w , x , y and z is the largest?

I. The average of w , x , y and z is 25.

II. The numbers w , x and y are each less than 24.

5. What is the average weight of the three new members who are recently included into the team?

I. The average weight of the team increases by 20 kg.

II. The three new men substitute earlier members whose weights are 64 kg, 75 kg and 66 kg.

6. The total of the present ages of A , B , C and D is 96 years. What is B 's present age? (SNAP, 2005)

I. The average age of A , B and D is 20 years.

II. The average age of C and D is 25 years.

7. What is the average age of children in the class?

(N.M.A.T., 2006)